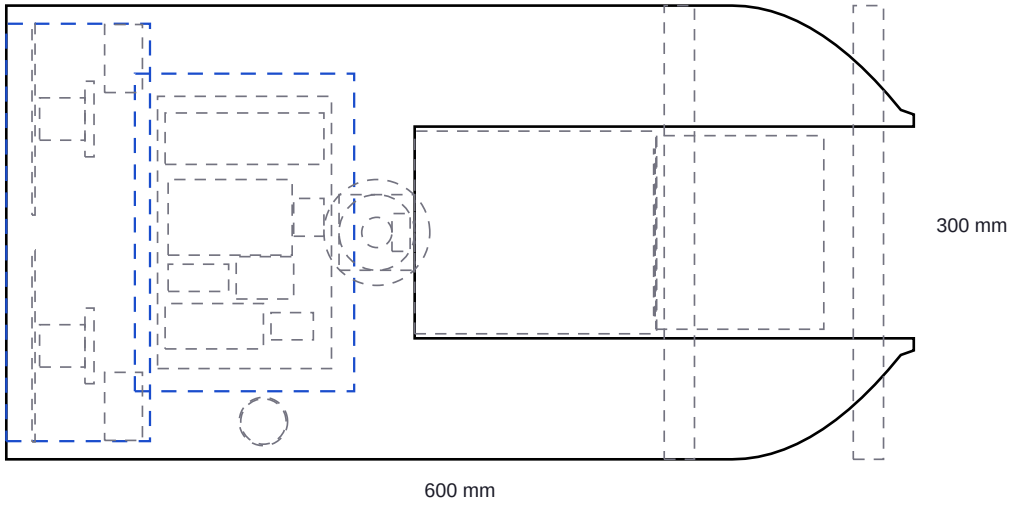
<- must measure 50 mm

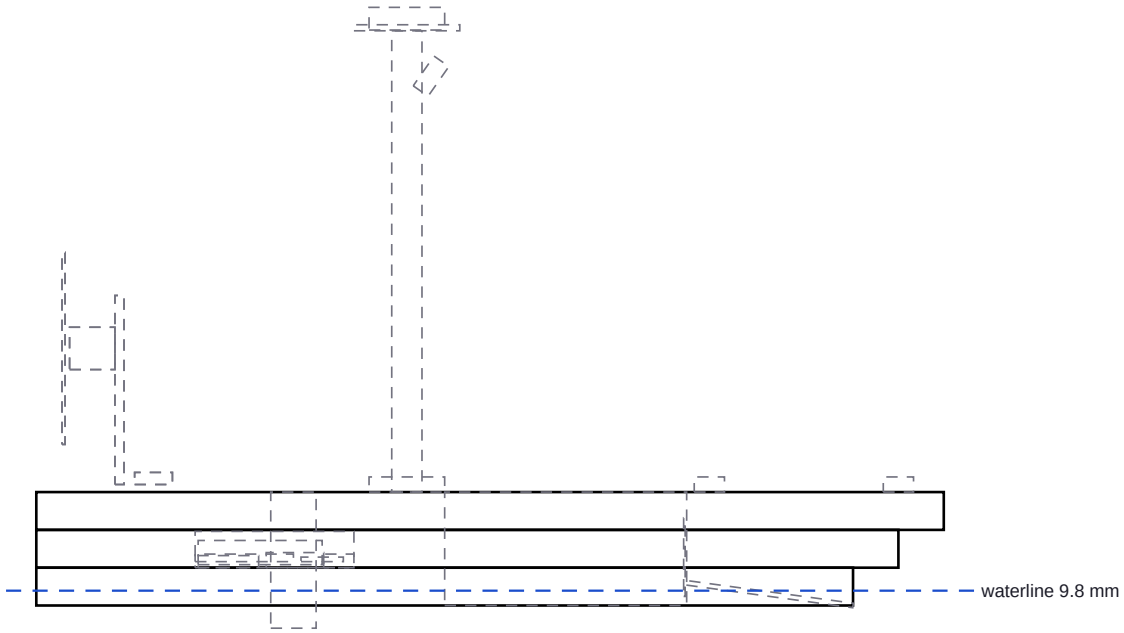
VARUNA U-Hull Block

600 x 300 mm thermocol boat | Easiest to build

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 75 mm tall, 396 mm to GPS top
Weight	1144 g empty, 1451 g with 307 g of plastic
Draft	9.8 mm empty, 12.4 mm full
Reserve buoyancy	6.7 kg before the deck goes under
Stability (GM)	833 mm: very stable
Trim	bow 6.2 mm higher than stern empty; slide the battery to level
Collection	1.54 L basket
Camera	340 mm above water, tilted 35 deg down
Battery	LiPo 3S 2200 mAh (24 Wh), about 0.8 h
Thermocol	3 sheets 1000 x 500, or 1 sheet 2 x 3 ft (610 x 915)

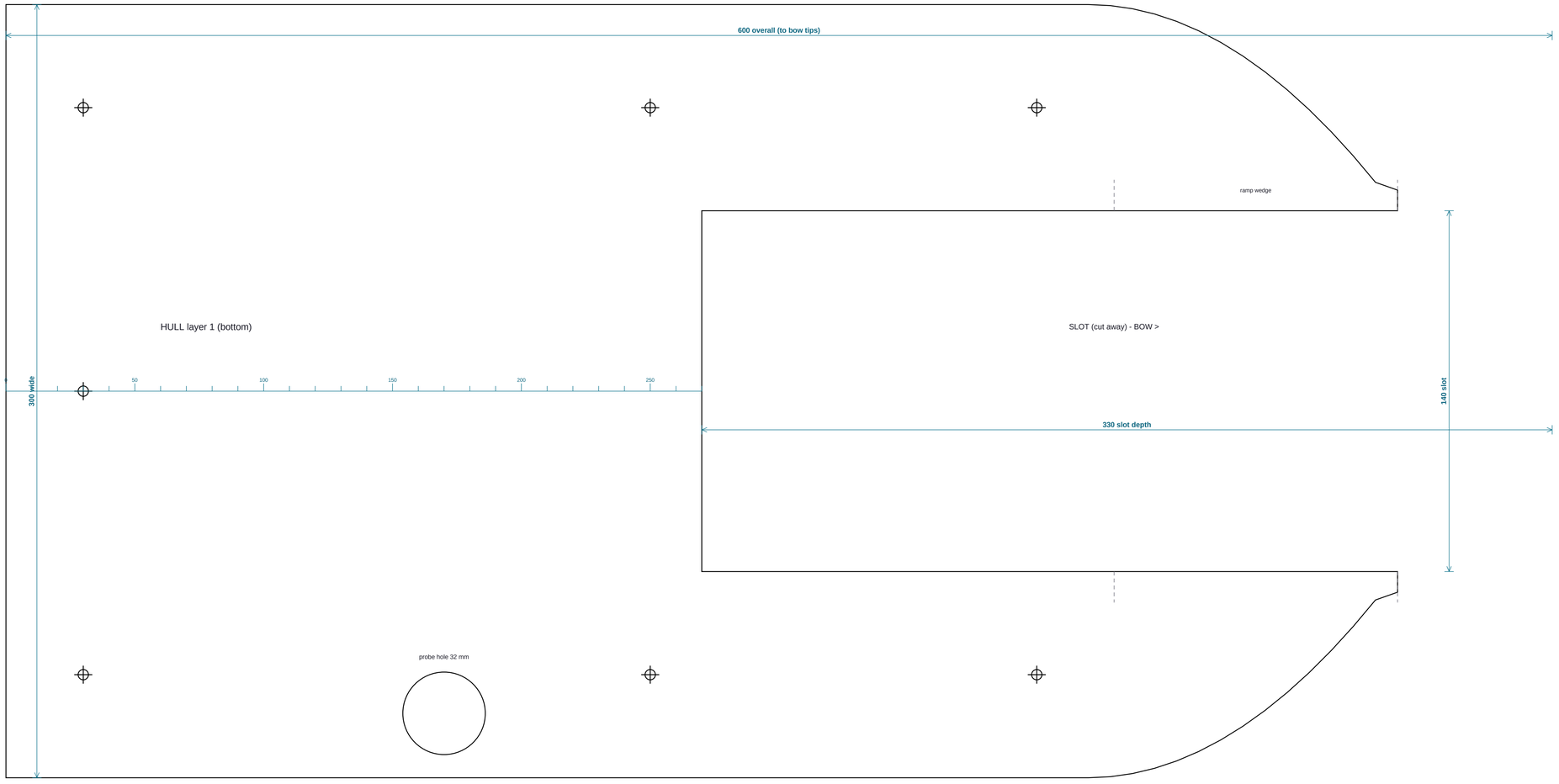
How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers

VARUNA U-Hull Block | Hull layer 1 (bottom) | 25 mm thermocol | A1, print at 100%

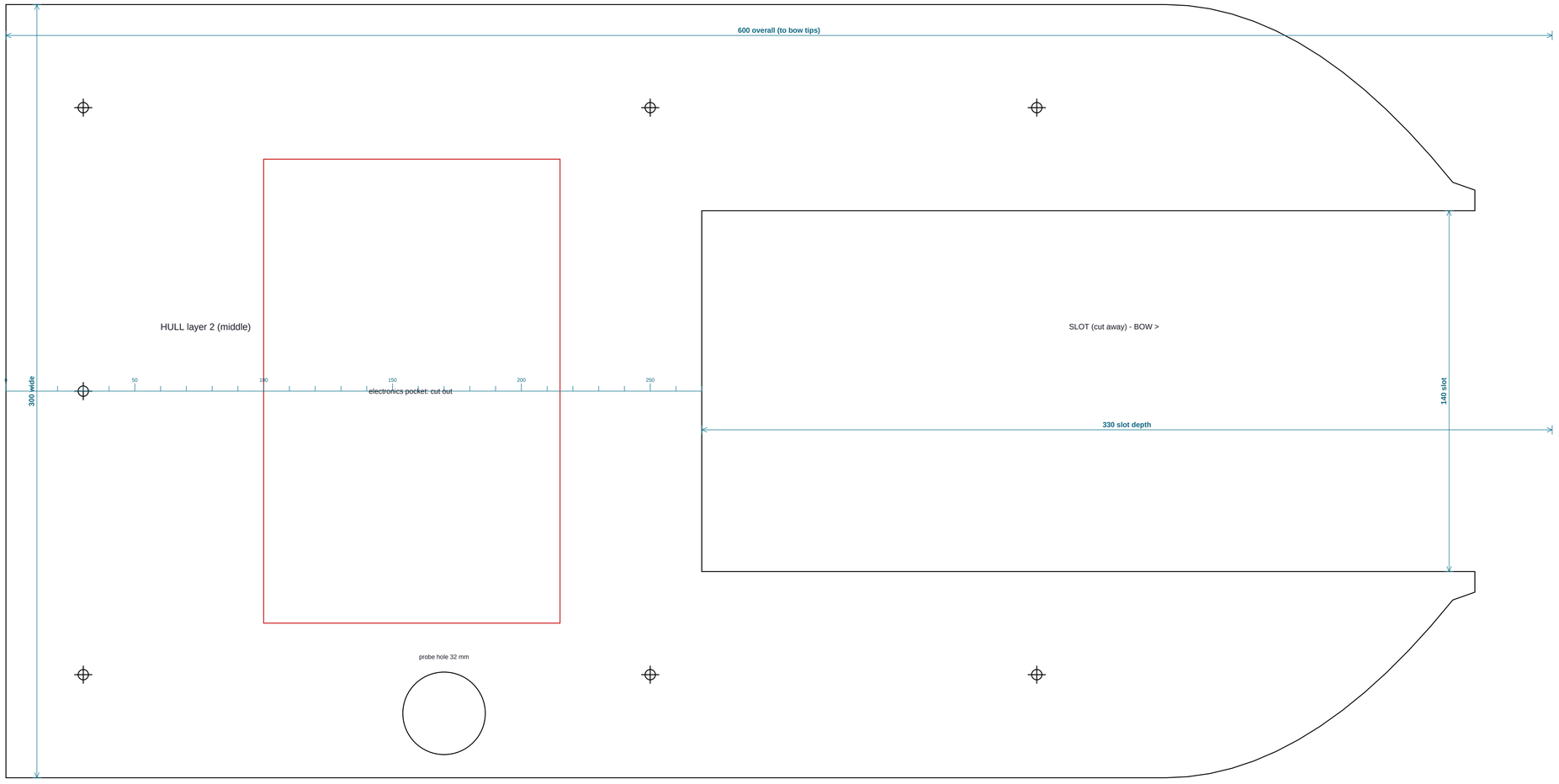
Push skewers through the dots to line up the layers.



<- must measure 100 mm

VARUNA U-Hull Block | Hull layer 2 (middle) | 25 mm thermocol | A1, print at 100%

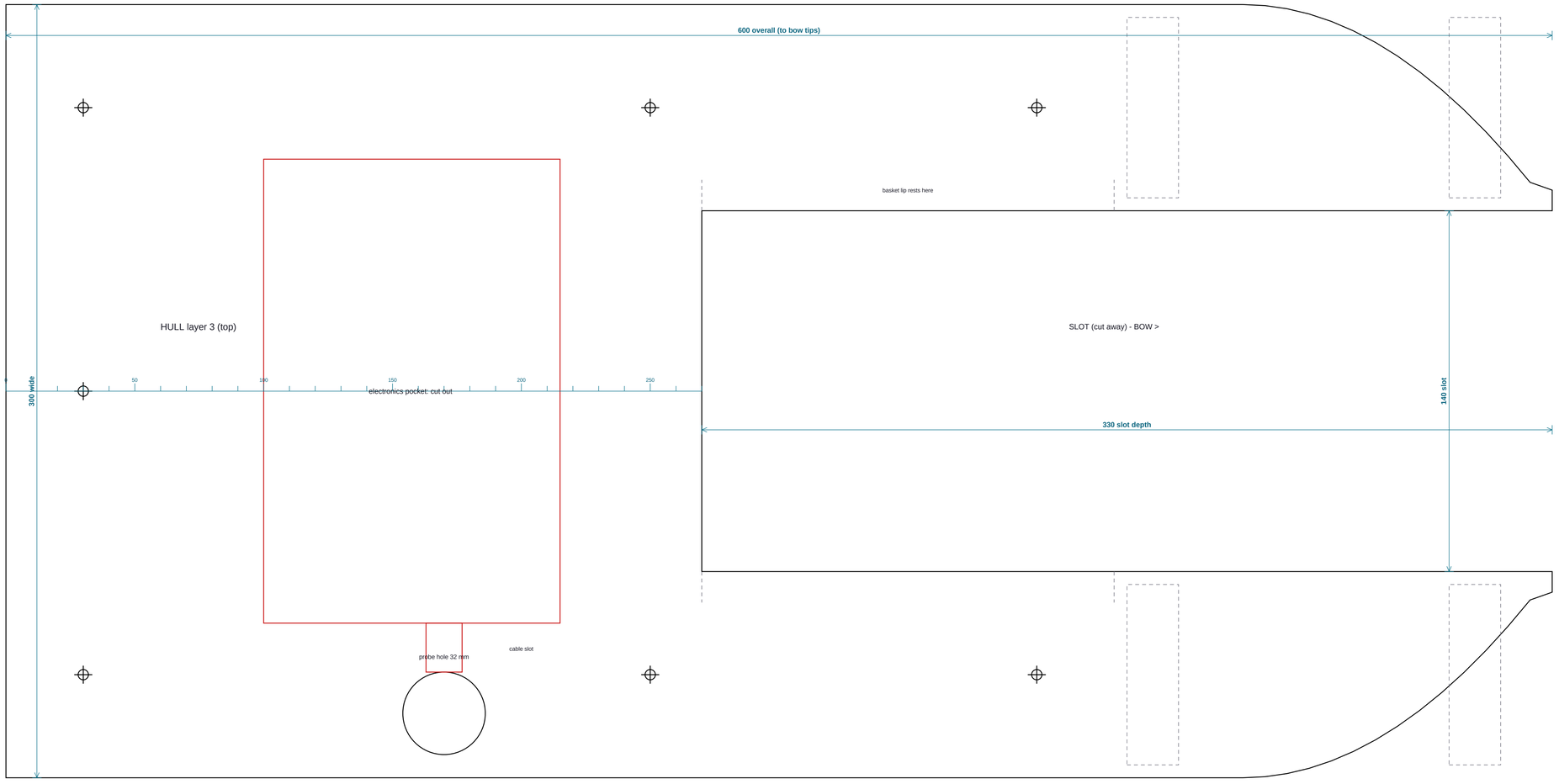
Push skewers through the dots to line up the layers.



<- must measure 100 mm

VARUNA U-Hull Block | Hull layer 3 (top) | 25 mm thermocol | A1, print at 100%

Push skewers through the dots to line up the layers.



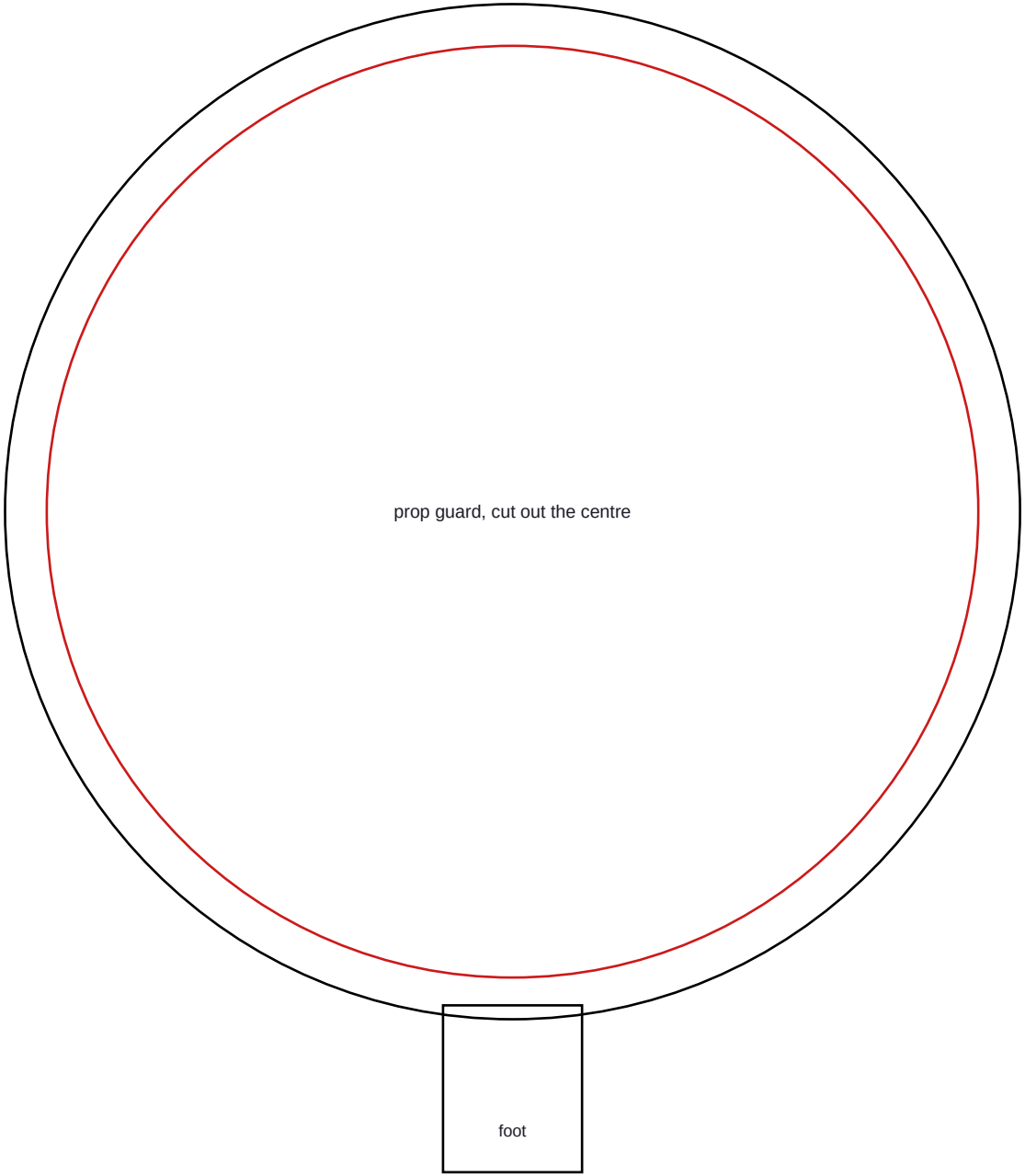
<- must measure 100 mm

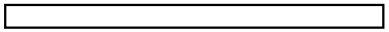
VARUNA U-Hull Block | small parts 1 of 10

<- must measure 50 mm

Prop guard ring (left)

5 mm sunboard | 146 x 168 mm

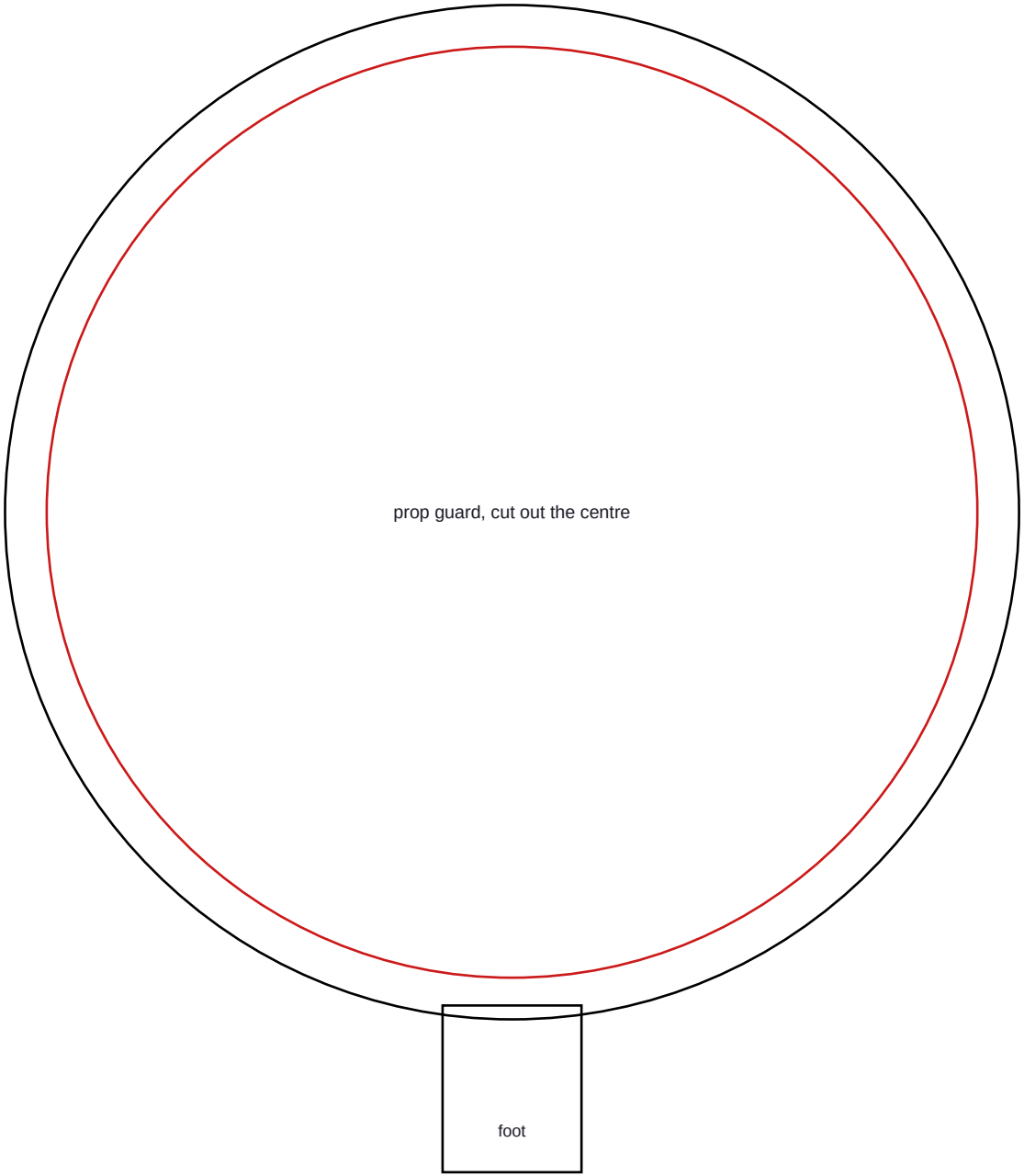




<- must measure 50 mm

Prop guard ring (right)

5 mm sunboard | 146 x 168 mm



<- must measure 50 mm

Hatch lid

5 mm sunboard | 210 x 145 mm

VARUNA U-Hull Block | small parts 4 of 10

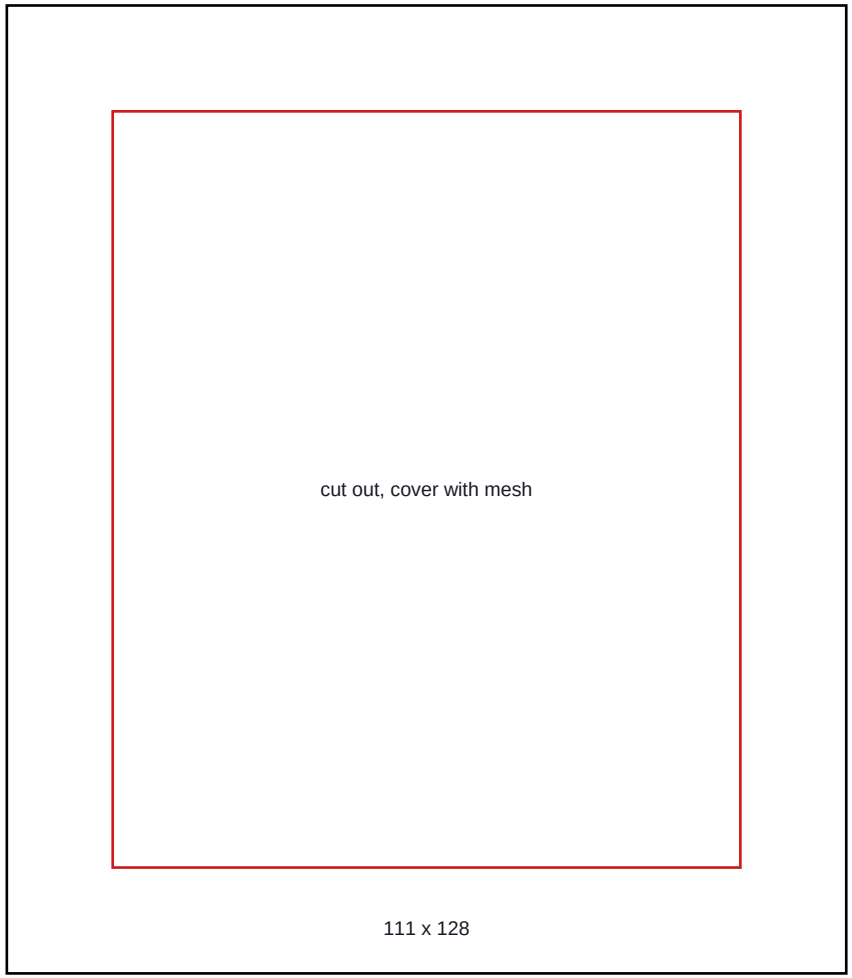
<- must measure 50 mm

Basket bottom

3 mm sunboard | 160 x 128 mm

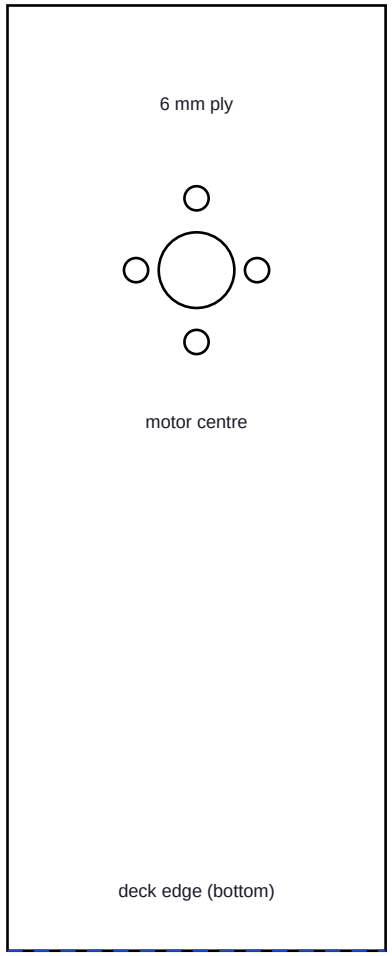
Intake ramp frame

3 mm sunboard | 111 x 128 mm

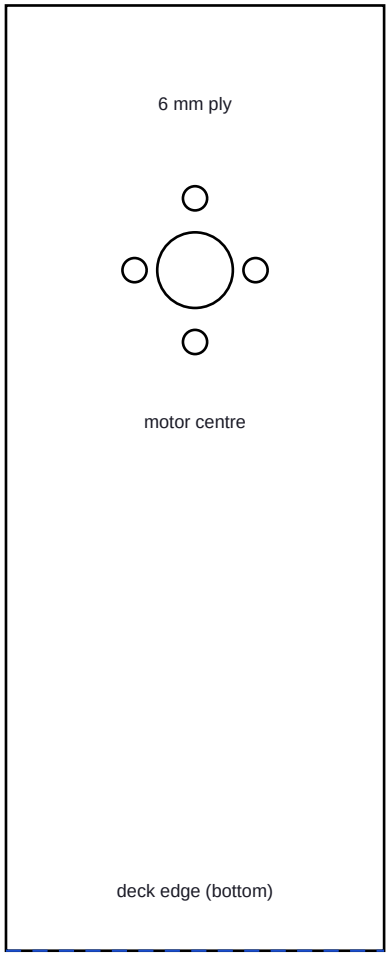


<- must measure 50 mm

Motor tower plate (left)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm



Motor tower plate (right)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm

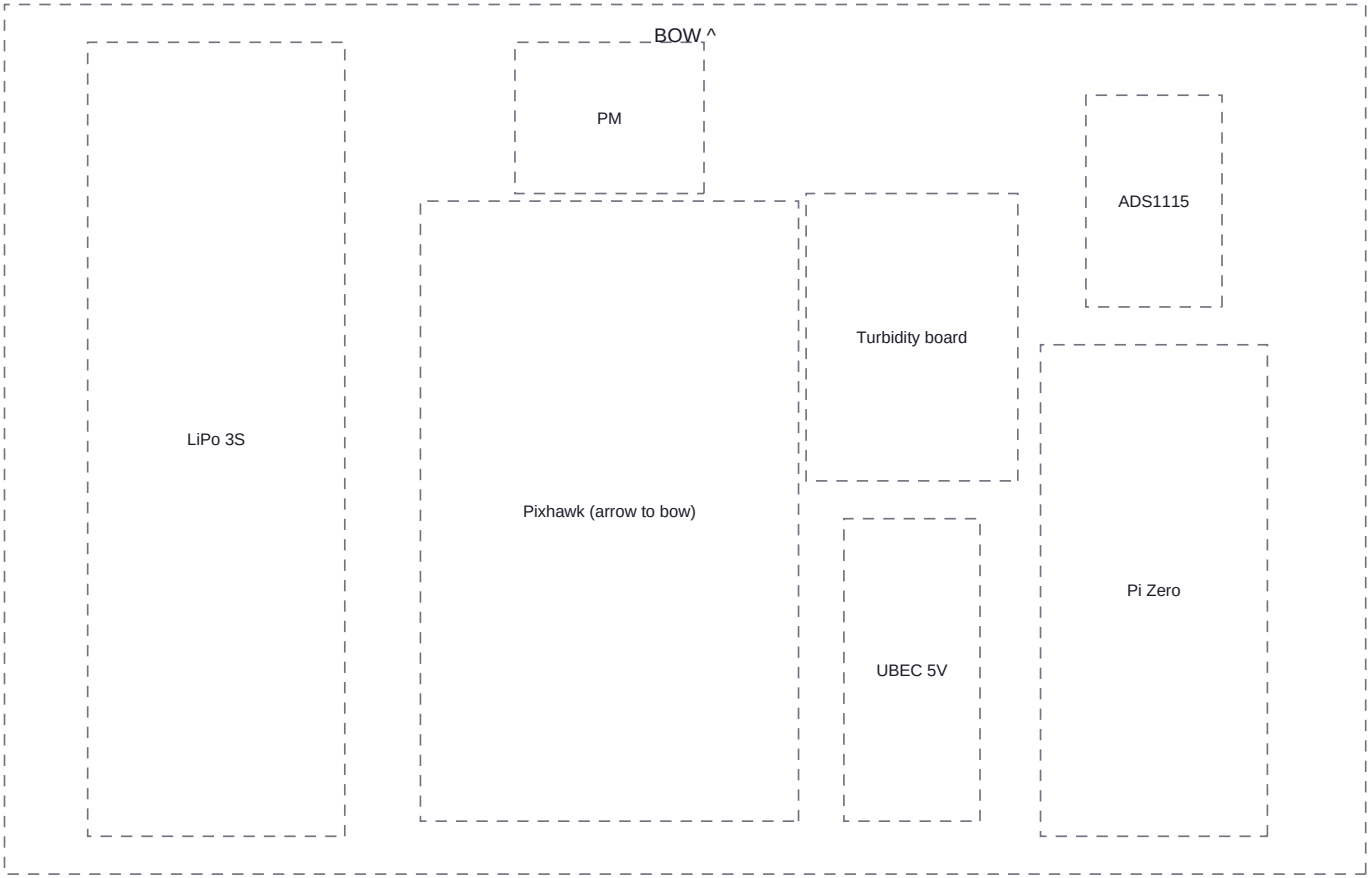


VARUNA U-Hull Block | small parts 6 of 10

<- must measure 50 mm

Pocket floor guide

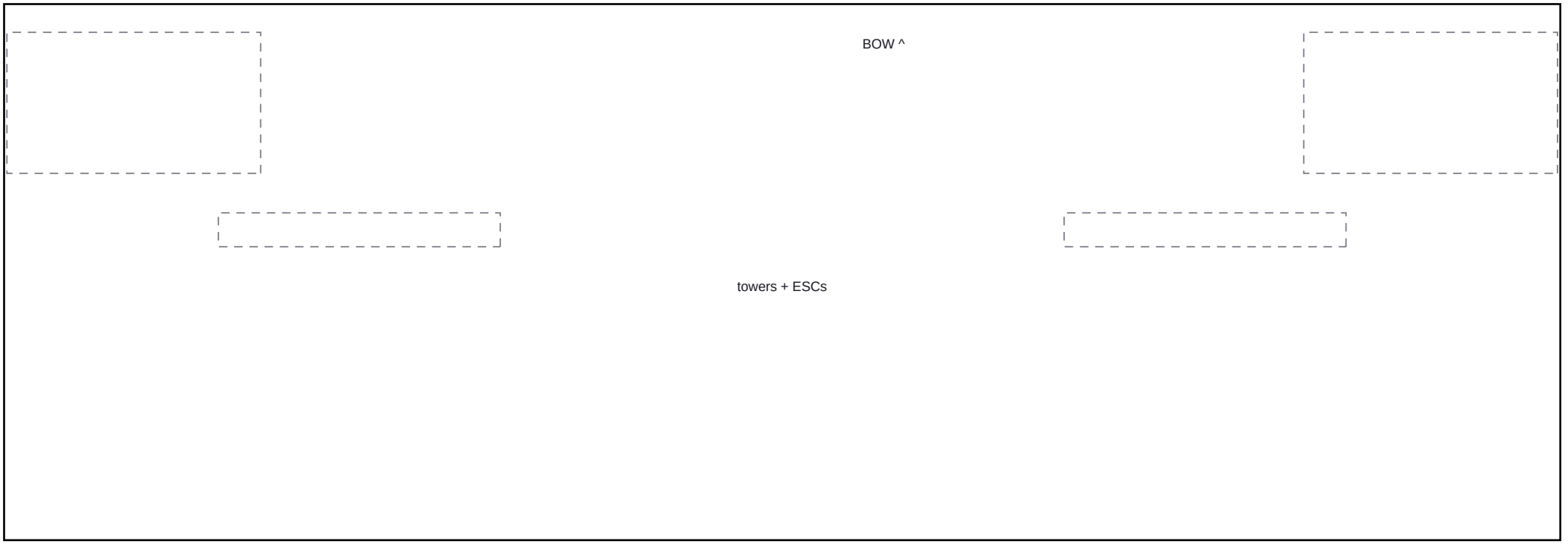
paper guide, glue inside the floor | 180 x 115 mm



<- must measure 50 mm

Stern plate

5 mm sunboard | 276 x 95 mm

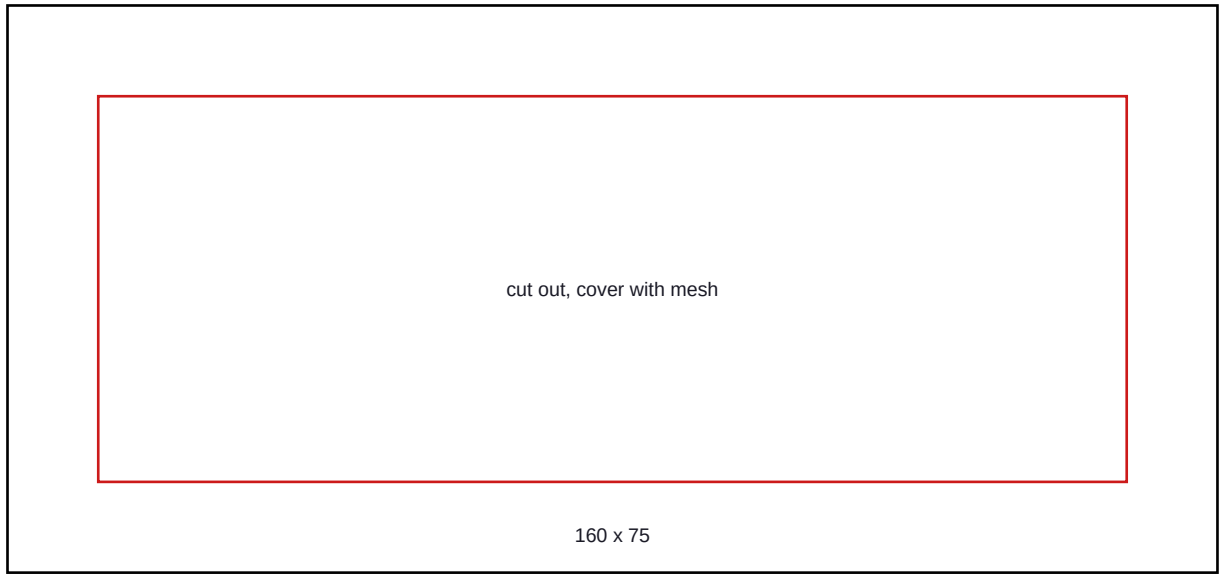


VARUNA U-Hull Block | small parts 8 of 10

<- must measure 50 mm

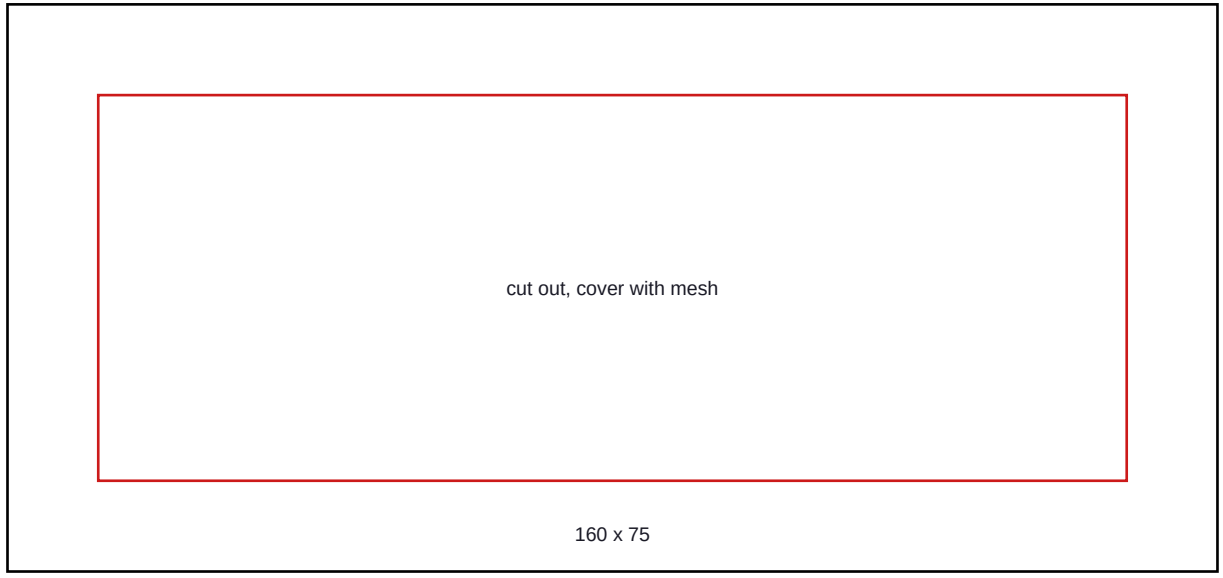
Basket side (left)

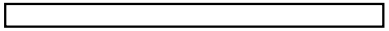
3 mm sunboard | 160 x 75 mm



Basket side (right)

3 mm sunboard | 160 x 75 mm

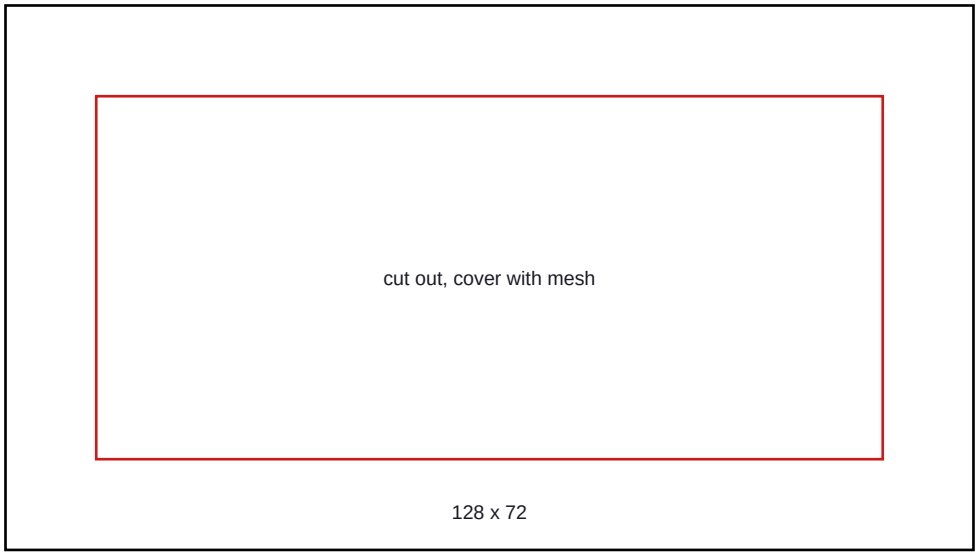




<- must measure 50 mm

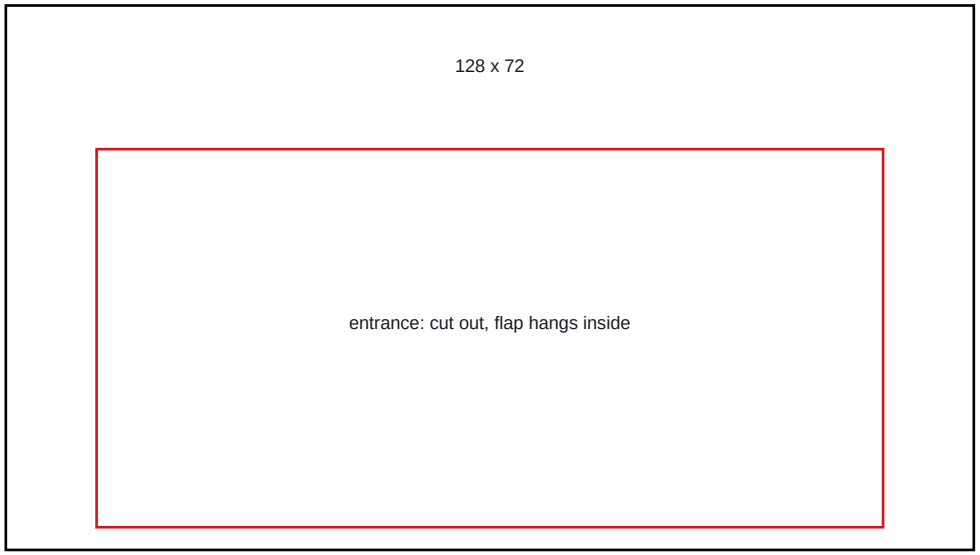
Basket back end

3 mm sunboard | 128 x 72 mm



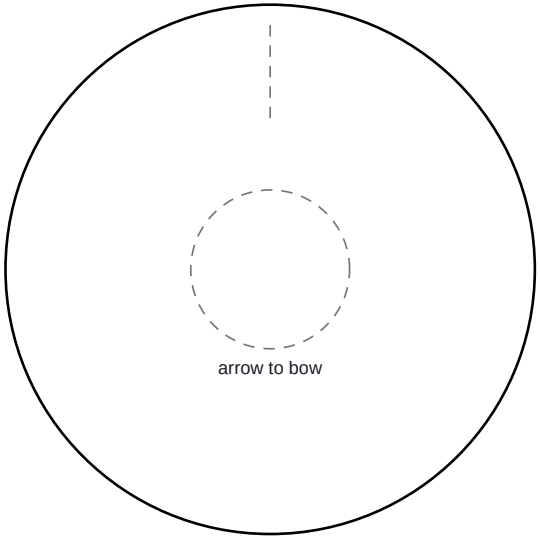
Basket front end (entrance)

3 mm sunboard | 128 x 72 mm



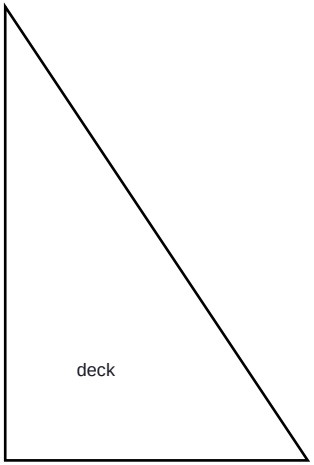
GPS disc

5 mm sunboard | 70 x 70 mm



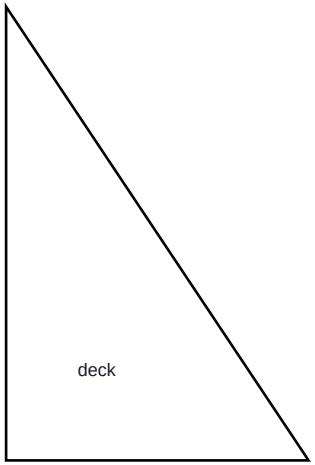
Tower gusset

5 mm sunboard | 40 x 60 mm



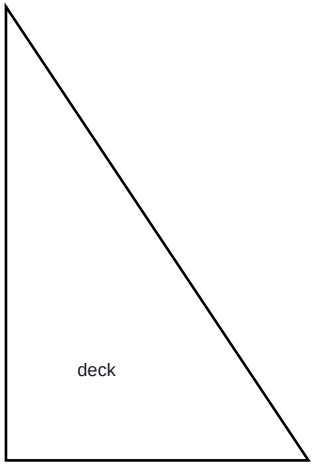
Tower gusset

5 mm sunboard | 40 x 60 mm



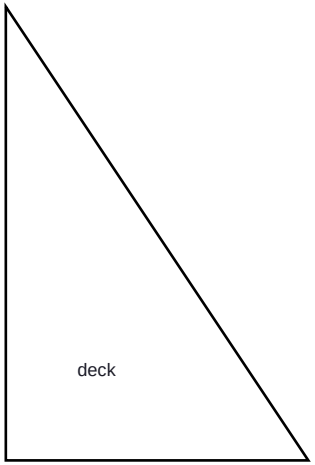
Tower gusset

5 mm sunboard | 40 x 60 mm

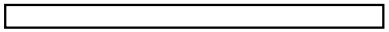


Tower gusset

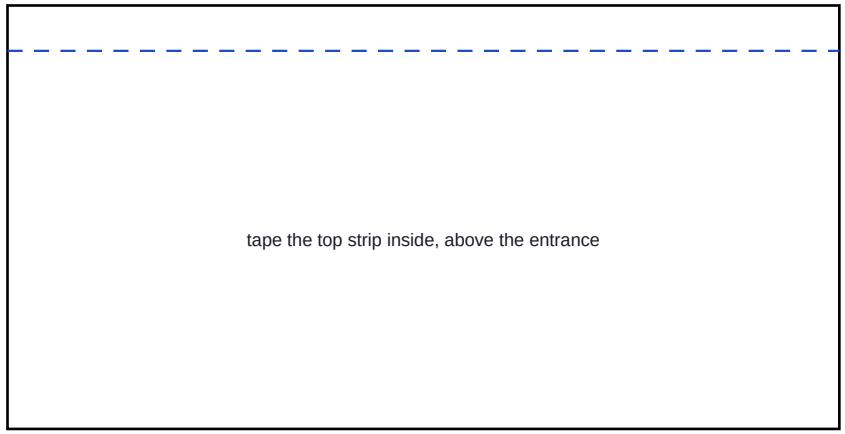
5 mm sunboard | 40 x 60 mm



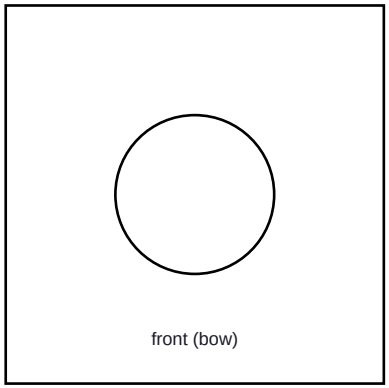
VARUNA U-Hull Block | small parts 10 of 10



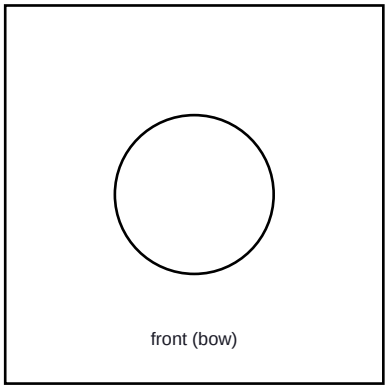
One-way flap
PET sheet from a bottle / transparency | 110 x 56 mm



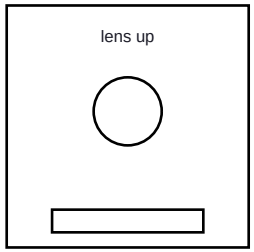
Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



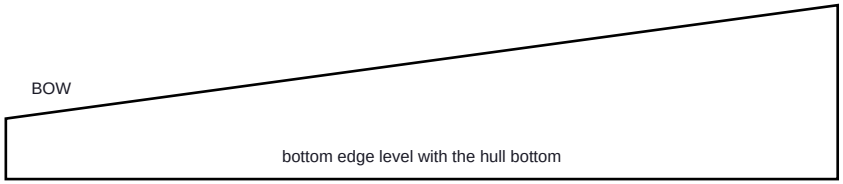
Camera plate
3 mm sunboard | 32 x 32 mm



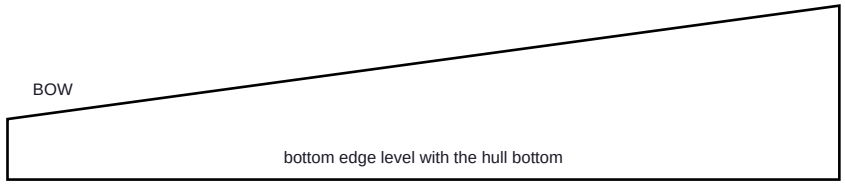
Basket lip (left)
3 mm sunboard | 160 x 28 mm

Basket lip (right)
3 mm sunboard | 160 x 28 mm

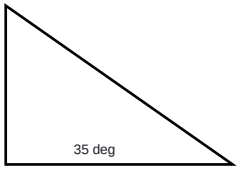
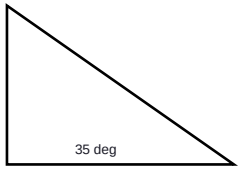
Ramp support wedge
5 mm sunboard (cut 2) | 110 x 23 mm



Ramp support wedge
5 mm sunboard (cut 2) | 110 x 23 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm